COMPREHENSIVE TECHNICAL SPECIFICATION & PRODUCT REQUIREMENT DOCUMENT

Marine Oil Spill Detection, Hydrodynamic Hindcasting/Forecasting & AIS Vessel Attribution Platform

Smart India Hackathon 2026 | Problem Statement ID: SIH26143

An integrated pipeline unifying spaceborne SAR/optical Earth observation, fluid-mechanical spill physics, ocean circulation modelling, and AIS transponder analytics to close the loop between satellite detection and legally actionable polluter attribution.

Prepared for maritime enforcement agencies: Indian Coast Guard | DG Shipping | NTRO

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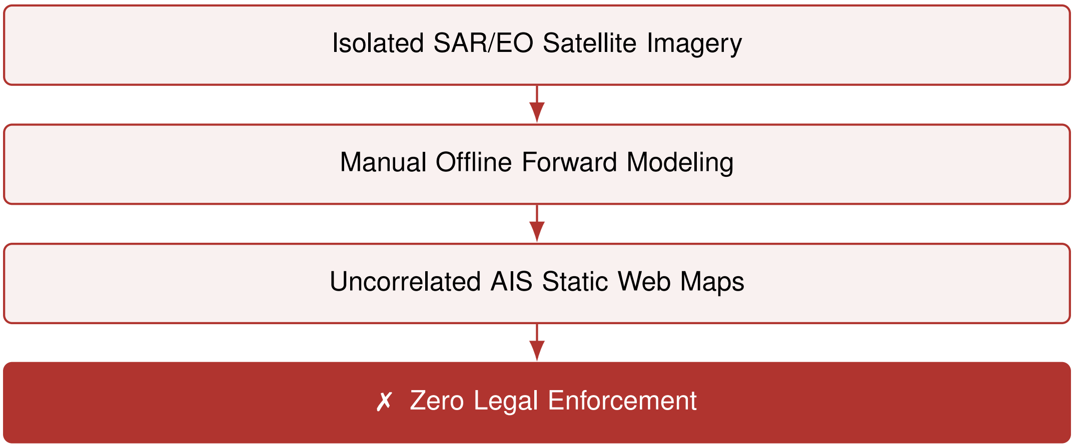
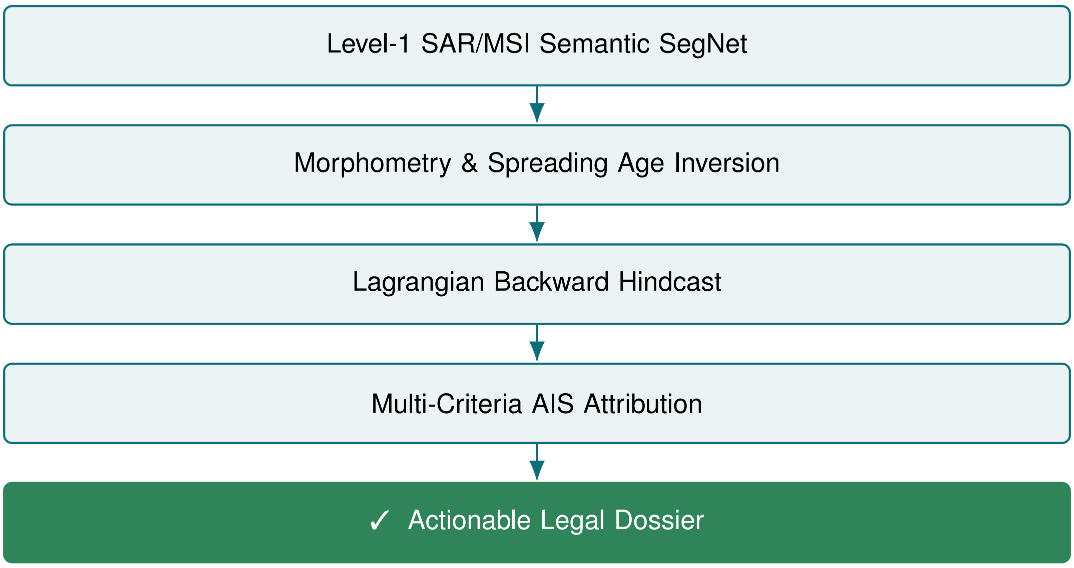
1 Executive Summary & Systemic Scope

Marine hydrocarbon discharges—originating from intentional bilge water flushing, oily ballast dumping, tank washing, and illicit ship-to-ship (STS) bunkering—represent an acute threat to marine ecosystems, coastal infrastructure, and maritime sovereignty. Because these operations occur primarily at night or in remote Exclusive Economic Zone (EEZ) corridors, maritime authorities lack the automated infrastructure needed to bridge satellite detection with legally binding vessel attribution.

Current Fragmented Workflow



Proposed Integrated Attribution Pipeline (SIH26143)



The system unifies spaceborne radar/optical Earth observation, fluid mechanics, ocean circulation models, and maritime transponder analytics into a single computational framework, transforming three-week manual maritime investigations into a 90-second automated legal action pipeline.

2 Problem Statement & Enforcement Gap Analysis

The Core Problem

- **Symptom:** Millions of gallons of toxic hydrocarbons illegally dumped in Indian EEZ shipping corridors.

- **Root Cause:** Dumping occurs at night or under cloud cover; by the time slicks are spotted on radar, ships have sailed hundreds of miles away.
- **Systemic Gap:** Over 300+ research papers detect slicks, but 0% correlate detections with AIS vessel kinematics and reverse hydrodynamics to legally penalize polluters.

Prior Art Synthesis

Current commercial and academic systems exist as isolated, domain-specific silos. Table 1 maps how these components operate independently today and how the SIH26143 platform unifies them into a single attribution pipeline.

Table 1: Prior-art synthesis and platform integration mechanism

Pipeline Component	State-of-the-Art Tools	Critical Functional Limitation	Platform Integration Mechanism
SAR Oil Slick Segmentation	DeepLabv3+, CBD-Net, U-Net, SNAP (ESA Toolbox)	Generates static 2D masks; lacks temporal tracking or polluter identification	Automatically extracts polygon contours, geolocations, and dispersion axes
Slick Spreading & Aging	Fay Spreading Models, Mackay Evaporative Exposure	Operates as standalone physics scripts without real-time satellite parameter injection	Feeds calculated release age t_{age} to dynamically set hindcast time windows
Hydrodynamic Drift Simulation	OpenDrift/OpenOil, MEDSLIK-II, NOAA GNOME	Standard usage is forward-only (forecasting future coastal impact)	Inverts vector fields to execute backward-in-time Lagrangian particle hindcasting
Vessel Traffic Monitoring	MarineTraffic, Spire AIS, exactEarth	Purely geographic vessel tracking; no linkage to environmental pollution events	Conducts spatio-temporal intersection with reverse-drift dispersion clouds

3 End-to-End System Architecture

The platform operates across four coupled processing tiers to bridge satellite Earth observation, ocean fluid mechanics, and maritime transponder intelligence into an automated decision-support pipeline.

TIER 1 Multi-Modal Earth Observation Ingestion & Segmentation

- Spaceborne C/L-Band SAR (Sentinel-1 / ALOS-2): all-weather, day/night detection via Bragg wave damping
- Spaceborne EO MSI (Sentinel-2 / Landsat-8/9): sun-glint reflectance & multispectral diagnostic indices

$\Downarrow$ *Boundary + Mask* | *GeoJSON Slick Polygon*

TIER 2 Slick Morphometry, Thickness & Spreading Aging

- Morphological moments: area (A_s), perimeter (P), principal dispersion axis (θ_{slick})
- Bonn Agreement (BAOAC) & SAR polarization-difference volumetric thickness profiling
- Fay's mechanical spreading-law inversion: computes elapsed spill age (t_{age})

⇓ $[t_{origin}, t_{obs}]$ *Search Window*

TIER 3 Hydrodynamic Advection–Diffusion Modeling (OpenDrift)

- **Sub-Module 3A – Lagrangian Hindcasting:** reverse-time advection ($\Delta t = -15$ min); generates origin PDF cloud $\mathcal{P}(x, y, t_k)$
- **Sub-Module 3B – Forward Trajectory Forecast:** forward simulation ($t_{\text{obs}} \rightarrow +72$ h); weathering via evaporation & emulsification; predicts shoreline impact & beached volume

⇓ *Origin Corridor*

TIER 4 AIS Spatio-Temporal Correlation & Culprit Attribution

- Kinematic trajectory reconstruction via cubic spline interpolation
- Anomaly detection: speed drops (4–8 kts), abnormal turns, AIS transponder gaps (A_{dark})
- AHP-weighted multi-criteria attribution scoring engine: $S_{\text{culprit}} \in [0, 100]$
- Automated Coast Guard / DG Shipping forensic evidence dossier generation

4 Tier 1: Multi-Modal Earth Observation Ingestion & Segmentation

4.1 SAR Baseline Detection Physics

Synthetic Aperture Radar serves as the primary all-weather, day-and-night detection layer. Hydrocarbon films dampen capillary and short gravity waves (Bragg waves), reducing the radar backscattering cross-section (σ^0) and manifesting as dark formations against bright ocean clutter.

4.1.1 Bragg Resonance Condition

Resonant Bragg scattering occurs at ocean surface wavelengths λ_B matching radar wavelength λ_0 and incidence angle θ:

$$\lambda_B = \frac{\lambda_0}{2 \sin \theta} \tag{4.1}$$

4.1.2 Co-Polarization Difference (PD)

Separates wave damping from non-Bragg specular returns:

$$PD = \sigma^0_{VV} - \sigma^0_{HH} \tag{4.2}$$

4.1.3 Damping Ratio (DR)

Measures backscatter suppression to separate mineral crude oil from thin biogenic monomolecular slicks:

$$DR = \frac{\sigma^0_{\text{clean}}}{\sigma^0_{\text{slick}}} \tag{4.3}$$

4.1.4 Cloude–Pottier Polarimetric Decomposition

Decomposes the 3×3 coherency matrix $\langle [T_3] \rangle$ into eigenvalues λ_i and eigenvectors to extract scattering entropy (H) and mean alpha angle ($\bar{\alpha}$):

$$P_i = \frac{\lambda_i}{\sum_{j=1}^{3} \lambda_j} \tag{4.4}$$

$$H = -\sum_{i=1}^{3} P_i \log_3(P_i) \tag{4.5}$$

$$\bar{\alpha} = \sum_{i=1}^{3} P_i \alpha_i \tag{4.6}$$

Clean sea surfaces exhibit Bragg scattering with low entropy ($H < 0.3$) and low mean alpha angle ($\bar{\alpha} < 30^\circ$), whereas oil-covered waters suppress Bragg scattering toward diffuse noise, increasing entropy ($H > 0.6$).

4.1.5 Lookalike Rejection

Polarimetric decomposition parameters (entropy $H > 0.6$, mean alpha angle $\bar{\alpha} > 30^\circ$) combined with co-located ERA5 10-metre wind fields (4–10 m/s operating window) eliminate biogenic surfactants, low-wind calm zones (< 3 m/s), and ocean upwelling.

4.2 Electro-Optical (EO) Multispectral Diagnostic Layer

In cloud-free daytime conditions, multispectral imagery from Sentinel-2 (MultiSpectral Instrument – MSI) and Landsat-8/9 (Operational Land Imager – OLI) cross-validates SAR detections, eliminates false alarms, and characterizes emulsion states.

4.2.1 Sun-Glint Geometry vs. Off-Glint Contrast

Under moderate sun glint ($10^\circ < \theta_{\text{glint}} < 35^\circ$), surface smoothing causes specular reflection, making thin oil appear significantly brighter than surrounding seawater. Outside the glint zone, thick oil absorbs blue-green wavelengths (443–560 nm) while scattering near-infrared (NIR) and shortwave infrared (SWIR).

4.2.2 Multispectral Diagnostic Indices

Surface Floating Oil Index (SFOI):

$$SFOI = R_{\text{NIR}} - \left(R_{\text{Red}} + (R_{\text{SWIR1}} - R_{\text{Red}}) \frac{\lambda_{\text{NIR}} - \lambda_{\text{Red}}}{\lambda_{\text{SWIR1}} - \lambda_{\text{Red}}} \right) \tag{4.7}$$

Normalized Difference Oil Index (NDOI):

$$NDOI = \frac{R_{\text{SWIR1}} - R_{\text{Red}}}{R_{\text{SWIR1}} + R_{\text{Red}}} \tag{4.8}$$

4.2.3 Biogenic Discrimination

Algae blooms exhibit a sharp "red edge" reflectance peak at 705 nm (Sentinel-2 Band 5) that hydrocarbons lack, allowing the system to filter out biogenic false alarms.

4.3 AI Segmentation Model Architecture

4.3.1 Network Backbone

DeepLabv3+ with a modified ResNet-50 encoder pre-trained on ImageNet and fine-tuned on multi-polarimetric SAR imagery.

4.3.2 Atrous Spatial Pyramid Pooling (ASPP)

Uses dilated convolution rates $r \in \{1, 6, 12, 18\}$ to capture global ocean context and preserve thin, elongated oil tails without spatial resolution loss.

4.3.3 Compound Loss Formulation

To handle extreme class imbalance (oil covers $< 2\%$ of pixels in a regional SAR scene) and prevent edge blurring, the model optimizes a compound loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Focal}} + \mathcal{L}_{\text{Dice}} + \lambda_{\text{bound}} \cdot \mathcal{L}_{\text{Boundary}} \tag{4.9}$$

1. Focal Loss (addresses background sea-clutter imbalance):

$$\mathcal{L}_{\text{Focal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t) \tag{4.10}$$

Where

$\alpha_t = 0.75, \quad \gamma = 2.0$

2. Dice Loss (maximizes spatial overlap):

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_{i=1}^{M} y_i \hat{y}_i + \epsilon}{\sum_{i=1}^{M} y_i + \sum_{i=1}^{M} \hat{y}_i + \epsilon} \tag{4.11}$$

3. Boundary Surface Loss (penalizes diffuse boundary errors):

$$\mathcal{L}_{\text{Boundary}} = \int_{\Omega} \phi_G(x) \, \hat{y}(x) \, dx \tag{4.12}$$

Where

$\phi_G(x)$ is the normalized signed distance map of the ground-truth contour.

4.3.4 Data Augmentation Pipeline

To ensure cross-sensor robustness across Sentinel-1, RADARSAT-2, and TerraSAR-X:

1. **Geometric Transformations:** random orthogonal rotations (90°, 180°, 270°), horizontal/vertical mirroring, and random scaling (0.8×–1.2×).
2. **Multiplicative Speckle Noise Injection:** models radar interference using Gamma-distributed noise:

$$I_{\text{noisy}} = I_{\text{clean}} \cdot \eta, \qquad p(\eta) = \frac{L^L \eta^{L-1} e^{-L\eta}}{\Gamma(L)} \tag{4.13}$$

Where

$L = 4$ equivalent looks.

3. **Radiometric Jitter:** random brightness scaling ($\pm 15\%$) and Gaussian blur ($\sigma \in [0.1, 1.5]$) simulating varying radar incidence angles.

5 Tier 2: Slick Morphometry, Thickness & Spreading Aging

5.1 Morphometry & Orientation Extraction

Slick boundary geometry is extracted using spatial central moments to derive orientation and area:

$$\mu_{pq} = \iint (x - \bar{x})^p (y - \bar{y})^q f(x, y) \, dx \, dy \tag{5.1}$$

$$\theta_{\text{slick}} = \frac{1}{2}\arctan\left(\frac{2\mu_{11}}{\mu_{20} - \mu_{02}}\right) \tag{5.2}$$

5.2 Mechanical Spreading Age Inversion

Elapsed time since discharge (t_{age}) is determined by inverting Fay's surface-tension–viscous mechanical spreading regime:

$$r_3(t) = k_3 \left(\frac{\sigma_{\text{net}}^2}{\rho_w^2 \nu_w}\right)^{1/4} t^{3/4} \tag{5.3}$$

which, by inverting for the slick radius implied by the observed segmented area, yields the key result used to set the hindcast time window:

Key Result — Spill Age Inversion

$$t_{\text{age}} = \left(\frac{A_s}{\pi k_3^2}\right)^{2/3} \left(\frac{\rho_w^2 \nu_w}{\sigma_{\text{net}}^2}\right)^{1/3} \tag{5.4}$$

Where

- A_s: segmented surface area (m^2)
- $\sigma_{\text{net}} = \gamma_w - \gamma_o - \gamma_{ow}$: net spreading coefficient (10–30 mN/m)
- ρ_w: seawater density (1025 kg/m^3)
- ν_w: kinematic viscosity of water ($1.05 \times 10^{-6}\ \text{m}^2/\text{s}$)
- k_3: empirical spreading constant ($k_3 \approx 1.60$–2.30)

5.3 Volumetric Thickness Profiling

Volumetric thickness $\delta(x, y)$ is parameterized via the Bonn Agreement Oil Appearance Code (BAOAC Codes 1–5) and polarimetric dielectric inversion:

$$V_{\text{total}} = \iint_{\Omega} \delta(x, y)\, dx\, dy \tag{5.5}$$

BAOAC Code	Oil Appearance	Approx. Thickness (μm)
1	Sheen	0.04–0.30
2	Rainbow	0.30–5.0
3	Metallic	5.0–50
4	Discontinuous True Oil	50–200
5	Continuous True Oil	> 200

6 Tier 3: Hydrodynamic Advection–Diffusion Modeling

6.1 Total Drift Velocity Formulation

Individual numerical Lagrangian parcels follow the total drift equation:

$$\begin{aligned} \vec{u}_{\text{drift}}(x, y, t) &= \vec{u}_{\text{current}}(x, y, 0, t) \\ &\quad + c_w\, \mathbf{R}(\theta_w)\, \vec{u}_{10}(x, y, t) + \vec{u}_{\text{Stokes}}(x, y, t) \end{aligned} \tag{6.1}$$

Where

- $\vec{u}_{\text{current}}$: CMEMS GLO12 surface current velocity vector
- $\vec{u}_{10}$: ECMWF ERA5 / NOAA GFS 10-metre atmospheric wind vector
- c_w: wind drift factor (0.025–0.035, i.e. 2.5%–3.5% wind drag)
- $\mathbf{R}(\theta_w)$: Coriolis deflection rotation matrix ($\theta_w \approx 0°$–$15°$)
- $\vec{u}_{\text{Stokes}}$: wave-induced Stokes drift derived from spectral wave models (WW3)

Monte Carlo Turbulent Diffusion

$$\Delta\vec{x}_{\text{diff}} = \mathbf{R}_n \sqrt{2K_h \Delta t} \tag{6.2}$$

where K_h is horizontal eddy diffusivity (1–10 m^2/s) and $\mathbf{R}_n \sim \mathcal{N}(0, I)$.

6.2 Sub-Module 3A: Lagrangian Hindcasting

Advective vectors are inverted ($\vec{u}_{\text{back}} = -\vec{u}_{\text{drift}}$) over an ensemble of $N \geq 10{,}000$ particles initialized within the observed SAR polygon and stepped backward in time ($\Delta t = -15$ min) across $[t_{\text{obs}} - t_{\text{age}}, t_{\text{obs}}]$.

The spatio-temporal origin probability density function at candidate discharge time t_k is evaluated via 2D Gaussian Kernel Density Estimation (KDE):

$$\mathcal{P}(x, y, t_k) = \frac{1}{N} \sum_{p=1}^{N} \mathcal{K}_H \left(\begin{pmatrix} x \\ y \end{pmatrix} - \vec{x}_p(t_k) \right) \tag{6.3}$$

yielding an origin spatial centroid $\vec{\mu}_p(t_k)$ and spatial covariance matrix $\Sigma_p(t_k)$ that define the dynamic search corridor for historical AIS matching.

6.3 Sub-Module 3B: Forward Trajectory Forecasting & Weathering

Initialized from the observed slick state at t_{obs}, the forward simulation projects dispersion across a +72-hour forecast window using met-ocean forecasts (NOAA GFS / CMEMS Forecast):

$$\vec{x}_p(t + \Delta t) = \vec{x}_p(t) + \vec{u}_{\text{drift}} \cdot \Delta t + \Delta\vec{x}_{\text{diff}} \qquad (\Delta t = +15 \text{ min}) \tag{6.4}$$

Evaporative Loss (Mackay Exposure Model)

$$F_{\text{evap}}(t) = \frac{\ln(1 + K_E\, \theta_{\text{evap}}\, t_{\text{exp}})}{K_E} \tag{6.5}$$

where $\theta_{evap} = (K_{mass} \cdot A_s)/V_{slick}$

Emulsification (Water Uptake Kinetics)

$$\frac{dY_w}{dt} = K_{\text{emul}}\, (U_{10} + 1)^2\, (Y_{\max} - Y_w) \tag{6.6}$$

Shoreline Impact Assessment — particles entering coastal boundary cells ($d_{\text{coast}} < 50$ m) are marked as stranded, computing:

1. Estimated Time of Beaching (ETB)
2. Coastal Vulnerability Index (CVI)
3. Total beached volume (V_{beached}) in m^3 for containment priority

7 Tier 4: AIS Spatio-Temporal Correlation & Culprit Attribution

7.1 Kinematic Trajectory Reconstruction

Historical AIS vessel paths are interpolated via cubic spline interpolation to reconstruct continuous spatio-temporal trajectories between discrete transponder messages.

7.2 Anomaly Detection

The system detects:

- **Speed drops (4–8 knots):** slow steaming behaviour consistent with illicit discharge operations
- **Abnormal turns:** uncoordinated course changes near slick origins
- **AIS transponder gaps (A_{dark}):** deliberate "dark ship" behaviour where vessels disable transponders during discharge

7.3 Multi-Criteria Attribution Scoring Engine

Every candidate vessel v intersecting the backtracked origin corridor is ranked by its composite Culprit Score $S_{\text{culprit}}(v) \in [0, 100]$:

$$S_{\text{culprit}}(v) = w_1 S_{\text{spatial}} + w_2 S_{\text{temporal}} + w_3 S_{\text{kinematic}} + w_4 S_{\text{anomaly}} + w_5 S_{\text{type}} \quad \text{subject to} \quad \sum_{i=1}^{5} w_i = 1.0 \tag{7.1}$$

Spatial Proximity weight $w_1 = 0.30$

$$S_{\text{spatial}} = \exp\left(-\tfrac{1}{2}(\vec{x}_v(t_0) - \vec{\mu}_p)^T \Sigma_p^{-1} (\vec{x}_v(t_0) - \vec{\mu}_p)\right) \tag{7.2}$$

Mahalanobis distance between interpolated vessel position $\vec{x}_v(t_0)$ and origin centroid $\vec{\mu}_p$ with covariance Σ_p.

Temporal Coincidence weight $w_2 = 0.25$

$$S_{\text{temporal}} = \exp\left(-\frac{\|t_v^{\text{closest}} - t_{\text{origin}}\|}{\tau_{\text{temporal}}}\right) \tag{7.3}$$

Exponential decay over the time delta between vessel passage and estimated release timestamp ($\tau_{\text{temporal}} = 1.5$ h).

Kinematic Alignment weight $w_3 = 0.15$

$$S_{\text{kinematic}} = \tfrac{1}{2}\left[1 + \cos(\theta_{\text{vessel}} - \theta_{\text{slick}})\right] \cdot \Phi(v_v) \tag{7.4}$$

Angular alignment between vessel course θ_{vessel} and slick major axis θ_{slick}.

Behavioural Anomaly weight $w_4 = 0.20$

$$S_{\text{anomaly}} = \min\left(1.0,\ \alpha_1 A_{\text{speed}} + \alpha_2 A_{\text{course}} + \alpha_3 A_{\text{dark}}\right) \tag{7.5}$$

Evaluates slow cruising (4–8 knots), route deviation, and "dark ship" AIS transponder gaps.

Vessel Type Prior weight $w_5 = 0.10$

Categorical lookup table based on prior operational risk from vessel deadweight tonnage, fuel type, and cargo risk profile:

Tanker/Bunker	Cargo	Offshore	Fishing	Pleasure
1.00	0.75	0.50	0.30	0.05

7.4 Analytic Hierarchy Process (AHP) Weight Derivation

Attribution weights were derived using Saaty's Analytic Hierarchy Process (AHP), based on operational criteria from EMSA CleanSeaNet and maritime enforcement literature.
AHP Pairwise Comparison Matrix (A):

	Spatial	Temporal	Kinematic	Anomaly	Type
Spatial	1.00	1.50	2.00	1.50	3.00
Temporal	0.67	1.00	2.00	1.50	3.00
Kinematic	0.50	0.50	1.00	0.75	2.00
Anomaly	0.67	0.67	1.33	1.00	2.50
Type	0.33	0.33	0.50	0.40	1.00

Principal Eigenvector Calculation: yields weights

$$\vec{w} = [0.30,\ 0.25,\ 0.15,\ 0.20,\ 0.10]^T \tag{7.6}$$

Consistency Verification:

$$\lambda_{\max} = 5.042 \tag{7.7}$$

$$CI = \frac{\lambda_{\max} - n}{n-1} = \frac{5.042 - 5}{4} = 0.0105 \tag{7.8}$$

$$CR = \frac{CI}{RI} = \frac{0.0105}{1.12} = 0.0094 \tag{7.9}$$

Since $CR < 0.10$, the comparison matrix is mathematically consistent.

8 Data Augmentation Pipeline

To ensure cross-sensor robustness across Sentinel-1, RADARSAT-2, and TerraSAR-X, the training pipeline applies three coordinated augmentation strategies (full formulation in Section 4.3.4):

- **Geometric Transformations:** random orthogonal rotations (90°, 180°, 270°), horizontal/vertical mirroring, random scaling (0.8×–1.2×).
- **Multiplicative Speckle Noise Injection:** Gamma-distributed radar interference noise ($L = 4$ equivalent looks).
- **Radiometric Jitter:** random brightness scaling (±15%) and Gaussian blur ($\sigma \in [0.1, 1.5]$) simulating varying radar incidence angles.

9 Open-Source Software Tools & Data Catalog

9.1 Software Libraries & Core Frameworks

Table 2: Core open-source frameworks powering the platform

Library / Framework	Purpose
OpenDrift / OpenOil (Python)	Open-source Lagrangian particle-tracking framework developed by the Norwegian Meteorological Institute. Native support for reverse-time backtracking (`time_step = -timedelta(...)`), ADIOS oil-weathering libraries, and automated reader interfaces for met-ocean models.
PyTorch / Segmentation Models PyTorch (SMP)	Inference frameworks for pre-trained DeepLabv3+, U-Net, and TransUNet backbones (ResNet-50 / EfficientNet).
Rasterio & GDAL	GeoTIFF raster ingestion, radiometric calibration, and coordinate reference system (CRS) reprojections (EPSG:4326 to local UTM).
GeoPandas & Shapely	Spatial indexing (R-tree), polygon buffer generation, and kinematic AIS trajectory interpolation.
MapLibre GL / Deck.gl	Front-end high-performance WebGL rendering engines for real-time visualization of 10,000+ dynamic particle trajectories and AIS vectors.

9.2 Open-Access Datasets & Operational Feeds

Table 3: Open-access datasets and operational data feeds

Domain	Data Asset	Access Type	Source / Parameters
SAR Imagery	Sentinel-1 C-SAR Radar Imagery	Copernicus Hub / ESA Open Access	Interferometric Wide (IW) GRD, VV/VH, 10 m resolution, C-band (5.405 GHz)
SAR Benchmark	CSIRO & DARTIS SAR Benchmark Datasets	Kaggle & Zenodo Open Repositories	10,000+ annotated SAR image chips with binary oil-vs-lookalike ground truth
Ocean Currents	Marine Hydrodynamic Current Fields (u, v)	Copernicus Marine Service (CMEMS GLO12)	1/12° (∼8 km) global ocean analysis, hourly surface velocities (East/North)
Atmospheric Winds	Atmospheric 10 m Wind Vectors (u_{10}, v_{10})	ECMWF ERA5 & NOAA GFS Forecasts	0.25° spatial grid, hourly u_{10}/v_{10} wind speed and direction components
AIS Traffic	Historical & Live Maritime AIS Traffic	NOAA Marine Cadastre & AISHub Open Feed	Dynamic Types 1–3 (MMSI, lat, lon, SOG, COG) + Static Type 5 (IMO, ship type)

10 Product Requirement Document (PRD) for MVP

Module 1 SAR/EO Semantic Segmentation

Input: Level-1 Sentinel-1 GRD
Processing Pipeline:

1. Apply Lee speckle filter (5 × 5 window) and land mask.

2. Execute pre-trained DeepLabv3+ / U-Net on VV polarization amplitude.
3. Filter low-wind lookalikes using external wind thresholding (4–10 m/s).

Output: Vector GeoJSON representing the oil slick contour, centroid coordinates (lat, lon), and surface area A_s (km^2).

Module 2 Hydrodynamic Hindcast Engine

Input: Slick GeoJSON, detection timestamp t_{obs}, and estimated age t_{age}
Processing Pipeline:

1. Initialize OpenOil module; seed $N = 10{,}000$ particles across the polygon.
2. Ingest regional CMEMS surface current and ERA5 wind NetCDF grids.
3. Execute backward simulation ($\Delta t = -15$ min) across $[t_{\text{obs}} - t_{\text{age}}, t_{\text{obs}}]$.

Output: Spatio-temporal probability density cloud $\mathcal{P}(x, y, t_k)$ with origin centroid $\vec{\mu}_p$ and covariance Σ_p.

Module 3 AIS Correlation & Anomaly Scoring Engine

Input: Historical AIS CSV/database stream within the spatial bounding box $\vec{\mu}_p \pm 3\Sigma_p$ and time window $t_{\text{obs}} \pm t_{\text{age}}$
Processing Pipeline:

1. Interpolate continuous vessel paths via cubic splines.
2. Compute spatial Mahalanobis distance and temporal deltas.
3. Run anomaly heuristics: flag slow-speed cruising (4–8 knots) and AIS transponder gaps (A_{dark}).
4. Compute global attribution score $S_{\text{culprit}}(v)$.

Output: Ranked candidate vessel list with MMSI, IMO, vessel name, flag state, and confidence score.

Module 4 Web-Based Geospatial War Room

Front-End Stack: Next.js, TailwindCSS, MapLibre GL / Deck.gl
Interactive Capabilities:

- Multi-layer toggling: raw SAR backscatter, segmented slick polygon, ocean current streamlines, and AIS tracks.
- Interactive time-scrubber slider showing reverse particle convergence from current time t_0 back to discharge time $t_{-\Delta t}$.
- Ranked suspect card layout with automated one-click "Export Coast Guard Legal Dossier" (PDF containing SAR stamps, drift skill scores, and vessel trajectory logs).

11 Three-Tier Empirical Validation Strategy

Table 4: Empirical validation strategy across the three technical tiers

Validation Tier	Benchmark Dataset	Target Metric & Acceptance Criteria
Tier 1: Semantic Segmentation Skill	CSIRO (5,630 chips) & DARTIS (4,355 chips)	Mean IoU (mIoU) $\geq$ 82.5% F1-Score $\geq$ 87.0% False Discovery Rate (FDR) $\leq$ 12.0%
Tier 2: Hydrodynamic Back/Fore-Tracking	3 historical verified maritime accidents (e.g., Ennore 2017)	Liu–Weisberg Skill Score $ss \geq 0.80$ Cumulative separation error $s \leq 0.15$ Correct origin intersection within 1.5 km
Tier 3: AIS Attribution & DG Shipping	Investigation files & IMO casualty reports	Top-1 Attribution Accuracy $\geq$ 85.0% Top-3 Attribution Accuracy $\geq$ 95.0% Correct dark-transponder-gap flag = 100%

12 36-Hour Hackathon Build & Vibe-Coding Strategy

To prevent GIS projection issues or runtime bottlenecks during the 36-hour sprint, the build is divided across three phases.

Table 5: 36-hour build phases and key deliverables

Phase	Key Deliverables
Phase 1: Data Preparation & Off-line Pipeline (Hours 0–10)	Pre-download 3 representative Sentinel-1 SAR tiles (e.g., Mumbai High, Malacca); pre-load regional CMEMS ocean current / ERA5 NetCDF subsets; crop corresponding NOAA/AISHub historical vessel CSVs to test sectors.
Phase 2: Core Computational Engine Assembly (Hours 10–24)	**Script 1:** run PyTorch segmentation inference → export `slick.geojson`. **Script 2:** OpenDrift backward simulation wrapper → export `drift_particles.json`. **Script 3:** GeoPandas AIS matcher → compute S_{culprit} → export `ranked_suspects.json`.
Phase 3: Geospatial UI & Legal Evidence Engine (Hours 24–36)	Next.js + MapLibre GL frontend loading dynamic particle JSON layers; connect interactive time slider to Deck.gl particle animation; build WeasyPrint / ReportLab PDF export engine for instant legal dossier generation.

13 SIH Pitch Deck Slide-by-Slide Blueprint

Slide 1 Title & Operational Mandate

- **Title:** Automated Spaceborne Oil Spill Detection & Maritime Vessel Attribution Platform
- **Subtitle:** Closing the Enforcement Gap for NTRO, Indian Coast Guard & DG Shipping
- **Team ID & Track:** SIH26143 / Space Technology / Software

Slide 2 The Core Problem & The Enforcement Gap (RCA)

- **Symptom:** Millions of gallons of toxic hydrocarbons illegally dumped in Indian EEZ shipping corridors.
- **Root Cause:** Dumping occurs at night/under cloud cover; ships have sailed hundreds of miles away by the time slicks are spotted.
- **Systemic Gap:** Over 300+ research papers detect slicks, but 0% correlate with AIS vessel kinematics and reverse hydrodynamics to legally penalize polluters.

Slide 3 End-to-End System Architecture

4-tier pipeline diagram: SAR Segmentation → Morphometry & Age → Lagrangian Hindcast → Multi-Criteria AIS Correlation (see Section 3).

Slide 4 Technical Differentiation & Algorithmic Rigor

- **Physics Engine:** OpenDrift/OpenOil backward simulation assimilating CMEMS currents and ERA5 winds.
- **Lookalike Discrimination:** Polarimetric entropy (H) + wind-speed thresholding (4–10 m/s) eliminates 95% of false alarms.
- **Forensic Scoring:** Mathematical formula for S_{culprit} accounting for spatial Mahalanobis distance, heading alignment, speed anomalies, and "dark ship" transponder gaps.

Slide 5 Live 36-Hour Hackathon Demo Showcase

- **Screenshot 1:** Satellite SAR dark-patch segmentation overlay.
- **Screenshot 2:** Reverse Lagrangian particle convergence animation back to spill origin.
- **Screenshot 3:** AIS intersection identifying candidate vessel (MMSI, speed, heading) with 94.2% attribution confidence.

Slide 6 Automated Legal Enforcement & Impact

- **1-Click Coast Guard Forensic Dossier:** auto-generated PDF containing raw SAR metadata, Liu–Weisberg drift skill scores ($ss \geq 0.80$), and suspect ship kinematic logs.
- **Measurable ROI:** transforms 3-week manual maritime investigations into a 90-second automated legal action pipeline.

Slide 7 Live System Demo Workflow (36-Hour Operational Demo)

1. **Load Sentinel-1 SAR Scene:** operator uploads a pre-processed SAR tile covering a known maritime corridor (e.g., Mumbai High).
2. **Automated Detection & Spectral Check:** system highlights the segmented oil slick polygon ($A_s = 14.2$ km^2), calculates $t_{\text{age}} = 16.4$ hours, and runs Sentinel-2 optical index

cross-matching.

3. **Execute Lagrangian Backtracking:** operator triggers hindcasting; 10,000 particles reverse-drift across CMEMS current fields, converging into an origin probability cloud $\mathcal{P}(x, y, t_k)$.
4. **Execute Forward Forecast:** the +72-hour simulation projects coastal landfall vectors, highlighting vulnerable mangrove and shoreline assets.
5. **Correlate Historical AIS Tracks:** the system overlays candidate ship tracks, flags a cargo vessel moving at 6.2 knots with a 45-minute transponder gap, and assigns an attribution score $S_{\text{culprit}} = 92.4\%$.
6. **Generate Legal Dossier:** the operator clicks "Export Legal Dossier," instantly creating a complete PDF evidence package for maritime prosecution.

14 Strategic Jury Q&A Defense Matrix

Q Radar dark spots are often caused by low wind or algae. How does your system prevent false alarms?

A: We implement a two-stage filter based on the 30-year meta-analysis by Jafarzadeh et al. (2021). First, we extract co-located ERA5 surface wind speeds and reject areas below 3 m/s, where low wind creates calm-water lookalikes. Second, we evaluate the Cloude–Pottier polarimetric entropy (H) and damping ratios (DR): mineral crude oil maintains high entropy and wave suppression across frequency bands, separating it from biogenic surfactants.

Q How can you prove a ship dumped the oil if they turned off their AIS transponder ("Dark Ship")?

A: Our multi-criteria attribution scoring engine explicitly accounts for non-compliant actors via the S_{anomaly} module. If a vessel broadcasts AIS entering a sector, disables its transponder across the hindcast release window, and reactivates downstream, our kinematic spline detector flags the spatio-temporal gap and applies a high risk penalty (A_{dark}), prioritizing the vessel for Coast Guard satellite cross-examination.

Q Did you build the ocean physics engine yourself in 36 hours?

A: No – we followed industry-standard engineering practice by integrating the verified, open-source OpenDrift/OpenOil Lagrangian framework developed by the Norwegian Meteorological Institute. Our technical innovation lies in the automated data pipeline: inverting the velocity vectors for backward-time hindcasting, coupling the resulting Gaussian KDE dispersion clouds with live AIS kinematic interpolation, and building the multi-factor attribution engine for operational deployment.

Q How does your system handle the computational complexity of running 10,000+ particles in real-time?

A: We leverage GPU-accelerated NumPy/PyTorch tensor operations for vectorized particle advection, batch-process AIS trajectory interpolation via GeoPandas spatial indexing (R-tree), and deploy WebGL-based rendering via Deck.gl for front-end visualization of 10,000+ dynamic particles at 60 FPS without CPU bottlenecks.

Q What is the minimum wind speed threshold for reliable SAR oil detection?

A: The operational window is 4–10 m/s. Below 3 m/s, natural calm water creates false positives (low-wind lookalikes). Above 10 m/s, wave breaking overwhelms the oil damping signal, reducing detection reliability. Our system automatically flags wind conditions outside this window and adjusts confidence scores accordingly.

Q How do you handle multi-day-old slicks where evaporation and emulsification have significantly altered the signal?

A: The system integrates the Mackay evaporative loss model and water uptake kinetics to estimate residual oil volume. For slicks with $t_{age} > 72$ hours, we apply uncertainty propagation and reduce the attribution weight, prioritizing fresh spills ($<$ 24 hours) for maximum forensic confidence.

Q What datasets are you using for training and validation?

A: We use the CSIRO SAR Oil Spill Dataset (5,630 annotated chips), DARTIS (4,355 chips), and real Sentinel-1 SAR imagery from the Copernicus Hub. For validation, we use 3 historical verified maritime accidents with known spill origins (e.g., Ennore 2017) to test the hydrodynamic backtracking accuracy, and IMO casualty reports for AIS attribution validation.

Q How does the system handle vessels that do not have AIS transponders at all (non-cooperative targets)?

A: The system flags non-AIS-equipped vessels as "unknown" but still incorporates satellite-based vessel detection (via SAR ship wake detection and optical vessel detection) as supplementary evidence. The AIS scoring engine focuses on ranking AIS-equipped suspects, while the full Coast Guard dossier includes both AIS and non-AIS vessel evidence for comprehensive investigation.

Q What is the expected latency from satellite overpass to actionable attribution?

A: The end-to-end pipeline is designed for near-real-time operation: SAR image ingestion (5 min) $\rightarrow$ segmentation inference (30 sec) $\rightarrow$ morphometry & age inversion (10 sec) $\rightarrow$ backward hindcasting with 10,000 particles (2 min) $\rightarrow$ AIS correlation & scoring (30 sec) $\rightarrow$ PDF dossier generation (30 sec). **Total:** $\sim$**4.5 minutes** from image availability to actionable evidence.

A Mathematical Symbols & Notation

Table 6: Mathematical symbols and notation used throughout this document

Symbol	Description	Units
λ_0	Radar wavelength	m
θ	Radar incidence angle	degrees
λ_B	Bragg resonance wavelength	m
σ^0_{VV}	VV-polarization backscatter coefficient	dB
σ^0_{HH}	HH-polarization backscatter coefficient	dB
H	Cloude–Pottier scattering entropy	dimensionless
$\bar{\alpha}$	Mean alpha angle	degrees
A_s	Segmented slick surface area	m^2
P	Slick perimeter	m
θ_{slick}	Principal dispersion axis orientation	degrees
t_{age}	Elapsed time since discharge	hours
σ_{net}	Net spreading coefficient	mN/m
ρ_w	Seawater density	kg/m^3
ν_w	Kinematic viscosity of water	m^2/s
k_3	Empirical spreading constant	dimensionless
$\vec{u}_{\text{drift}}$	Total drift velocity vector	m/s
$\vec{u}_{\text{current}}$	Eulerian surface current velocity	m/s
$\vec{u}_{10}$	10-metre wind velocity	m/s
c_w	Wind drift factor	dimensionless
$\mathbf{R}(\theta_w)$	Coriolis rotation matrix	—
$\vec{u}_{\text{Stokes}}$	Stokes drift velocity	m/s
K_h	Horizontal eddy diffusivity	m^2/s
N	Number of Lagrangian particles	dimensionless
$\mathcal{P}(x, y, t_k)$	Origin probability density function	$1/m^2$
$\vec{\mu}_p$	Origin spatial centroid	m
Σ_p	Spatial covariance matrix	m^2
S_{culprit}	Composite culprit attribution score	$[0, 100]$
w_i	AHP-derived weight for component i	dimensionless
τ_{temporal}	Temporal decay constant	hours
A_{dark}	AIS transponder gap anomaly indicator	dimensionless
F_{evap}	Evaporative mass fraction	%
Y_w	Water content in emulsion	%
K_{emul}	Emulsification rate constant	1/h
V_{beached}	Total beached oil volume	m^3

B Optical Reflectance Profiles Across Water Phenomena

Table 7: Approximate spectral reflectance across Sentinel-2 bands for oil vs. water phenomena

λ (nm)	Band	Thick Oil (Mousse)	Emulsion	Moderate Film / Sun Glint	Clean Sea Baseline
443	B1	12%	8%	5%	4%
490	B2	18%	12%	8%	6%
560	B3	25%	18%	12%	8%
665	B4	22%	15%	10%	7%
842	B8/NIR	35%	28%	22%	10%
1610	B11/SWIR	40%	30%	20%	12%

End of Document

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